Asif Jawad

Software Engineer @ Samsung Research

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Research Interests
My research interests broadly lie in computer vision and representation learning, including **3D perception, spatial reasoning, and world models**. I am particularly interested in developing efficient and compact visual representations, while also considering reliability and interpretability. I also hope to explore multimodal learning and its applications to embodied intelligence and biomedical vision.

Education

University of Dhaka Jan 2019 – Jan 2024
B.Sc. in Computer Science and Engineering **CGPA:** 3.58/4.00

> **Undergraduate Thesis:** *Utility-based Hypergraph Mining*
> This research formed the basis of the **ADMA 2026** publication *UHGMiner: A Framework and Algorithm for High-Utility Hypergraph Mining*.

Publications

1 RaPTGS: Render-Agnostic Post-Training Compression of 3D Gaussian Splatting 2026
***Asif Jawad**, K. M. Azwad Hossain*
ECCV 2026 | Paper | Poster

> 3D Gaussian Splatting enables real-time, high-fidelity rendering of novel viewpoints, but its large point-based representation creates substantial storage costs that limit distribution and deployment on resource-constrained devices. RaPTGS addresses the model-only setting, in which training images, camera poses, and the rendering pipeline are unavailable. The method scores Gaussian importance using geometric, spatial, and appearance cues, prunes redundant elements, and restores local coverage through training-free refinement. Spatial reordering, vector quantization, and entropy coding further compact the representation. Across standard benchmarks, RaPTGS achieves about $30\times$ compression while maintaining competitive visual quality, making trained assets easier to archive, share, and deploy.

2 UHGMiner: A Framework and Algorithm for High-Utility Hypergraph Mining 2026
***Asif Jawad**, Jawad Habib, Md. Tanvir Alam, Md. Fahim Arefin, Chowdhury Farhan Ahmed, Carson K. Leung*
ADMA 2026

> Hypergraphs model relationships among several entities at once, such as coauthor groups or product bundles. Existing mining methods largely prioritize frequency, even though the most frequent patterns are not necessarily the most valuable. UHGMiner formulates the search around utility, allowing it to identify patterns whose combined value exceeds a chosen threshold. Its HWU and REU pruning methods eliminate unpromising candidates before expensive pattern matching. Experimental results validate the efficiency of the approach and show that the discovered patterns offer greater practical value than frequency-based alternatives.

Work Experience

Software Engineer, Samsung R&D Institute Bangladesh Feb 2024 – Present

Physical AI Simulator *(Python, NVIDIA Isaac Sim, Unreal Engine)* Jul 2026 – Present

> Developing a simulation framework for multiple autonomous humans using NVIDIA Isaac Sim and Unreal Engine by integrating navigation, animation, and behavior systems. The resulting indoor environments support walking, sitting, and pick-and-place interactions, providing controllable scenarios for Physical AI and embodied AI research.

3D Content Editor *(Vulkan, C++)* Sep 2025 – Jun 2026

> Contributed to a mobile application for viewing and interacting with 3D reconstruction models, with a focus on performance-critical sorting components. Prototyped GPU-accelerated parallel key–value sorting using Vulkan compute shaders, validating a high-performance alternative to an existing CPU bottleneck. Also implemented a production-grade C++ version of a parallel linear-assignment sorting algorithm based on academic literature, translating the method into efficient application code.

Samsung Notes *(UWP, WinUI 3, C#, C++)* Feb 2024 – Jun 2026

Contributed to Samsung Notes across its UWP/C# implementation and subsequent transition to WinUI 3/C++. During the UWP phase, analyzed the startup path and resolved system-level bottlenecks, reducing launch time by 30%. Built a proof of concept for double ListView virtualization with on-demand item population to improve UI scalability for large datasets. Later, led the initial migration of core components from C# to C++ and developed a C++ SDK module to support the new architecture. Implemented key authoring and document workflows in WinUI 3, including text input and rendering, thumbnail generation, page previews, PDF insertion, clipboard operations, drag-and-drop support, and drawing capabilities.

Achievements

🎖️ **Honorable Mention** at the 45th ICPC World Finals Standings 2022

Represented the University of Dhaka at the 45th ICPC World Finals. Received an Honorable Mention at the championship round of ICPC, the world's oldest, largest, and most prestigious collegiate programming contest. The edition's published scale was over 59,000 contestants from more than 3,400 universities in 111 countries. Only 137 advanced to the World Finals. Standings

🥇 **Champion** at ICPC Dhaka Regional Standings | News 2020

🥈 **1st Runner-up** at ICPC Dhaka Regional Standings 2026

🥉 **2nd Runner-up** at ICPC Dhaka Regional Standings 2025

🥇 **Champion** at UIU Inter-University Programming Contest Standings | News 2025

🥈 **1st Runner-up** at BUET Inter-University Programming Contest Standings 2024

🥈 **1st Runner-up** at AUST Inter-University Programming Contest Standings 2025

🥇 **Champion** at Bangladesh Olympiad in Informatics News 2014

🥇 **Champion** at National High School Programming Contest News 2015

🥇 **Champion** at National High School Programming Contest News 2016

⭐ **Candidate Master** on Codeforces, ranking in the top 4% of users globally Profile

🏆 **763rd place** among 22,000+ participants in Meta Hacker Cup, within the top 3.5% Certificate 2024

Professional Recognition

Samsung Icon Award Award

Awarded in recognition of outstanding contributions to Samsung.

Samsung Software Competency Professional Certificate

Passed Samsung's global, time-constrained Professional assessment in algorithms and data structures.

Samsung Code Contest

Served as lead judge and problem author for two editions of the annual programming competition.

Technical Mentorship

Mentored colleagues across multiple years for Samsung's Internal Capability Certification.

Technical Skills

Programming Languages

C++, Python, C# and Java

Machine Learning & Scientific Computing

PyTorch, NumPy, SciPy, OpenCV

Graphics & Development Frameworks

Vulkan, NVIDIA Isaac Sim, Unreal Engine, WinUI 3, UWP

Application Development

React, Django, SDL2, Firebase, SQLite, Git